

SPECIAL EVENT: SHELBY JACOBS ROCKETRY

This special event will be run at the Jacksonville competition in memory of Shelby Jacobs (1935-2022). Mr. Jacobs was one of the first African American aerospace engineers and worked on rockets for the Mercury, Atlas, Jupiter, Thor, Apollo, and Space Shuttle programs. Mr. Jacobs was a huge fan of Florida Elementary Science Olympiad and the opportunities that it gives to students. The Florida First Coast Science Olympiad competition will host a water rockets competition in honor of Mr. Jacobs. This is a special event and will not count toward overall team standings. Winners will receive special awards provided by Mr. Jacobs's family.

Description: Prior to the competition, teams will design and build a rocket propelled by air pressure and water out of a 1-liter plastic beverage bottle. The students will create an Engineering Design Log (EDL) as they work through the design process and submit it to the event supervisor. The rocket that stays in the air for the longest time accompanied by the best EDL will win. This is a PRE-BUILT EVENT.

Number of Participants: 2

Impound: Yes

Construction:

1. Teams will bring their rockets to the testing site in a box labeled with their team number. The rocket must also have the team number printed on it using a permanent marker.
2. The rocket fuselage/body must be made from a single clear, traditional plastic 1-liter beverage bottle used to hold water and air pressure that propels the rocket when released.
3. The 1-liter bottle must have a neck/nozzle opening internal diameter of approximately 2.2 cm. Some water bottles have different diameters and heights and they cannot be accommodated on the launcher. Some also have extended necks, making it impossible to attach to the launcher.
4. The bottle itself must not be altered in any way (e.g. holes, scratches, increasing the volume, restricting the bottle's opening). A safety check may be done to make sure that the bottle has not had any structural alterations due to physical, thermal, or chemical changes.
5. The rocket must not contain or be subject to any of the following: metal, commercial model rocket parts including the parachute, explosives, gases (except air), electronic devices, elastic, remote control, chemicals or pyrotechnics, super glue (cyanoacrylate), or high temperature hot glue.
6. Fins, parachutes, and other items may be added to the outside of the bottle to increase the time aloft. Adhesives that may be used to attach components to the bottle are limited to tape and glue (only silicone, polyurethane-based, or others that do not damage the structural integrity of the bottle).
7. All parts and any changes must be attached at a point that is 5 cm or more above the level of the bottle's opening. No part of the rocket can intrude into this 5 cm zone.
8. The nose of the rocket must be rounded or blunt with a maximum curvature that matches that of a penny. It can be less curved than a penny (i.e. not as sharp) but it can't be more curved (sharper) than a penny.
9. The fully assembled rocket must not exceed 50 cm in height and 20 cm in width, in its prelaunch state..
10. Energy to propel the rocket must come only from the water and air pressure in the bottle. Other sources of potential or kinetic energy are not allowed.
11. Only plain water may be used in the rocket. No other material of any type may be put in the bottle or added to the water. A water level line may be marked on the bottle to aid in adding water.
12. Parts of the rocket may deploy during flight, but they must remain attached by a string/lanyard.
13. Rockets will be launched from an Aquapod launcher. (available on online)

Impound:

1. Teams will impound their water rocket(s) in a box or container clearly marked with the team number. Event Supervisor during the impound process.
2. Teams will submit the Engineering Design Log to the Event Supervisor during impound.
3. Teams may impound repair kits containing tools, spare parts, and extra parachutes in their rocket box.
4. Any parts found to be dangerous (ex. glass or metal), illegal (ex. commercially made rocket parts), or that prevent a rocket from fitting on the launch pad must be removed before the rocket can be launched. Rockets that are changed to meet the construction requirements may be penalized.
5. A safety check may be done to make sure that the bottle has not had any structural alterations due to physical, thermal, or chemical changes.
6. Rockets deemed unsafe according to the event supervisor's judgment will not be launched.

The Competition:

1. Two launches will be allowed. Different rockets may be used for each launch.
2. Teams must use the water, launch pad, and source of pressure provided by the event supervisors.
3. The team will add the desired amount of water to the rocket before each flight and may make alterations or repairs to rocket(s) between launches.
4. Outside assistance/coaching from the sidelines is not permitted and will result in the team being placed in Tier 2.
5. The event supervisors will supervise the pressurization of the rocket to 30 psi and the launching of it. Students must not begin any action without explicit direction of the event supervisor. Anyone within 5 meters of a pressurized rocket must wear eye protection. Contestants may not hold their rocket during pressurization.
6. One team member will pull the cord to launch the rocket. The student must call out a warning at launch such as "3,2,1 launch".
7. It will be the supervisors' decision whether the flight should be considered as unofficial due to variable weather conditions such as high winds or an error on the part of the Aquapod.
8. Timing begins when the rocket separates from the launcher.
9. Timing ends when any part of the rocket touches the ground, rests on an obstruction, or goes out of sight.

Engineering Design Log:

Teams will be asked to submit an Engineering Design Log ([Engineering-Design-Log.pdf](#)):

1. Teams will use the template provided to create their EDL.
 - a. For "Identify a need" students should use the challenge set forth for this event.
 - b. "Specify Requirements" should clearly include design requirements for this event.
2. The team may make multiple copies of pages as needed to show the progression of their design(s).
3. The event supervisor will use the rubric provided in the EDL to help evaluate.
4. Teams will need to add a cover to the EDL with the team number and event title marked clearly.

Scoring:

1. Highest score wins.
2. The times of both flights will be added together to achieve a total time score. This will be combined with the score from the Engineering Design Log.
3. Scores will be determined for the rocket by taking the average of the two event supervisors' times for the duration of each rocket's flight.
4. The watches will be started simultaneously at time of launch and stopped when the rocket touches the ground or any object in contact with the ground, trees, buildings, etc.

5. Flights of rockets whose parts do not remain attached together during the entire flight, or that cannot be changed to meet the construction requirements, will be ranked in Tier 2 and then by their time aloft, behind all flights of rockets without construction violations and whose parts remain attached.
6. Teams whose rockets cannot be launched for any reason will receive participation points only.
7. Ties will be broken using the quality of the Engineering Design Log.

Launcher Links:

[Oriental Trading Company](#)

[Alumigo](#)

[Altakorea](#)

Special Event

Shelby Jacobs Rocketry Scoresheet

Team Number: A_____

Team Name: _____

Student Names: _____

IMPOUND

Were all items impounded during the appropriate time?	N = Tier 3	
Is the device within the design parameters?	N= Tier 3	

LAUNCH

Did the students wear protective eye covering?	N= Tier 2	
Outside interference or coaching?	Y=Tier 2	

SCORING

Launch #1 Stopwatch 1: _____ Stopwatch 2: _____	Average time:
Launch #2 Stopwatch 1: _____ Stopwatch 2: _____	Average time:
Engineering Design Log Score (up to 20 points)	Score:
Total of the two average flight times: _____	Tier _____